A Physical Layer Authentication Mechanism for IoT Devices

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Abstract: When Internet of Things (IoT) nodes access the network through wireless channels, the network is vulnerable to spoofing attacks and the Sybil attack. However, the connection of massive devices in IoT makes it difficult to manage and distribute keys, thus limiting the application of traditional high-level authentication schemes. Compared with the high-level authentication scheme, the physical layer authentication scheme realizes the lightweight authentication of users by comparing the wireless channel characteristics of adjacent packets. However, traditional physical layer authentication schemes still adopt the one-to-one authentication method, which will consume numerous network resources in the face of large-scale IoT node access authentication. In order to realize the secure access authentication of IoT nodes and regional intrusion detection with low resource consumption, we propose a physical layer authentication mechanism based on convolution neural network (CNN), which uses the deep characteristics of channel state information (CSI) to identify sending nodes in different locations. Specifically, we obtain the instantaneous CSI data of IoT sending nodes at different locations in the pre-set area, and then feed them into CNN for training to procure a model for IoT node authentication. With its powerful ability of data analysis and feature extraction, CNN can extract deep Spatio-temporal environment features of CSI data and bind them with node identities. Accordingly, an authentication mechanism which can distinguish the identity types of IoT nodes located in different positions is established to authenticate the identity of unknown nodes when they break into the pre-set area. Experimental results show that this authentication mechanism can still achieve 94.7% authentication accuracy in the case of a low signal-to-noise ratio (SNR) of 0 dB, which means a significant improvement in authentication accuracy and robustness.

Keywords: IoT; CSI; CNN; access authentication; intrusion detection

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I. INTRODUCTION

With the rapid development of IoT, a massive amount of sensitive and confidential data is expected to be transmitted on the wireless channel. However, the security and privacy of wireless communications is always the main challenge because of the broadcast nature of wireless transmissions [1]. The openness of the wireless link in IoT leaves the network unprotected to attacks that forge legitimate user identities, such as spoofing attacks and the Sybil attack [2], when IoT nodes access the network through access points (APs). Specifically, attackers can either capture legitimate node identities and break the data integrity in the process of data transmission or claim a large number of false identities to send high-speed access request messages to AP, which leads to the unnecessary consumption of AP resources and network congestion. However, traditional high-level authentication schemes based on key negotiation and distribution, such as group key negotiation [3], the adaptive private key establishment [4], one-time key issuance [5] and so on, almost face the challenges of public key infrastructure deployment and the high complexity of cryptographic algorithms. In addition, numerous overhead and network latency caused by the management and distribution of keys occur as massive devices are connected. To solve the security threat caused by the attackers, an immediate and efficient lightweight authentication scheme is necessary to identify the unknown IoT sending node.

Compared with the traditional key-based authentication scheme, physical layer authentication does not need key distribution and management, which greatly reduces the authentication complexity. Moreover, it is difficult for attackers to simulate the physical layer properties of wireless channels. Thus physical layer authentication can be well used in the identity authentication of IoT sending nodes. Zhang et al. [6] proposed a physical layer authentication scheme embedded with Gaussian tags, which exploited weighted fractional Fourier transform to help verify the identity of IoT entities and prevent unauthorized access to information or services. For the incoherent large-scale single-input multi-output (MIMO) industrial IoT communication system, Liao et al. [7] realized the physical layer authentication by embedding the authentication signal into the message signal. Huang et al. [8] extracted the representative time and frequency characteristics from the receive signal strength (RSS) to characterize the radio propagation curve, thus realizing the authentication of IoT devices.

However, due to the error of channel estimation, these traditional physical layer authentication schemes are poor in robustness and easily affected by channel fluctuations. In recent years, machine learning has been gradually applied to combine with physical layer authentication schemes because of its strong data analysis ability and is proved to be robust, reliable, and accurate. Xiao L et al. [9] proposed a physical layer spoofing attack detection system based on Q-learning, which can obtain the optimal test threshold in spoofing attack detection in advance. Based on [9], Dyna-Q learning [10] is adopted to improve authentication efficiency. Although the method of threshold processing using machine learning has improved the accuracy, it still realizes authentication by the one-to-one method. That is, AP needs to authenticate each IoT sending node and seek the optimal threshold value individually, which will consume plenty of network resources and cause signaling congestion. So it does not suit the access authentication of large-scale IoT nodes well.

In order to enhance the secure access authentication of multiple IoT nodes in the area, we propose a novel physical layer authentication mechanism based on CNN. CNN is recognized for its ability on feature extraction and classification, while the access authentication can essentially be summed up to distinguish the identities of the IoT nodes, so CNN can be used as a good enabling tool to assist the implementation of physical layer authentication. The scheme we proposed can break away from the limitation of the authentication threshold, therefore realizing "threshold freedom". Specifically, our method collects the instantaneous CSI data in the IoT area as the training data of CNN, so as to extract the wireless environment features of the region as the identity of the IoT nodes, and establishes a robust authentication model. Once a foreign node invades and initiates an access request, the instantaneous CSI data of the foreign node will be fed into the trained authentication model, and the deep CSI features are extracted to determine the identity type of the unknown node. The simulation results show that the physical layer authentication mechanism based on CNN can achieve the authentication accuracy of 94.7% even under the condition of low SNR of 0 dB. Compared with the traditional physical layer authentication scheme, our method improves accuracy and robustness significantly.

II. SYSTEMS MODEL

In the actual communication scenario, multiple IoT legal nodes, such as sensor nodes, collect information and access the network layer through AP. The network layer carries out functions like data coding, high-level authentication, and data transmission. However, the openness of the wireless channel may make AP subjected to spoofing attacks and the Sybil attack when IoT nodes access AP, where spoofing attack nodes can capture the legitimate node identity and monitor or tamper the data in the process of data transmitting; Sybil nodes can claim multiple fake legitimate node identities, thus initiating high-speed access requests to AP, which will occupy and consume AP resources, resulting in network congestion.

The system model is shown in Figure 1. We consider an IoT scenario, where a single AP and multiple IoT legal nodes are located within a pre-set area. AP and legitimate nodes are artificially deployed and their locations are fixed. The whole area in the IoT scene is divided into grids of equal size, and the grids where

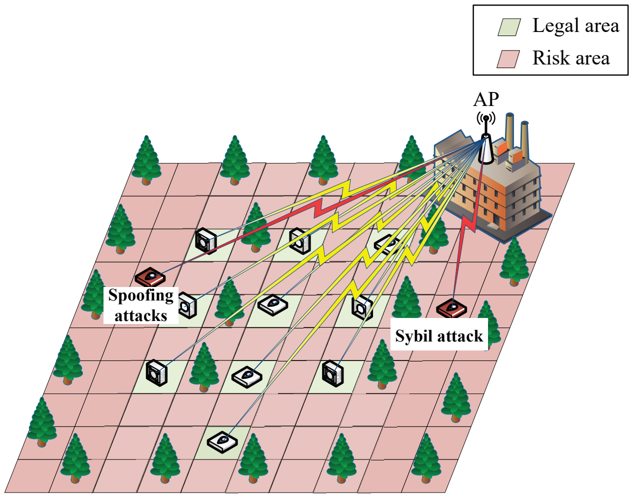


Figure 1. *System model.*

legitimate nodes located are marked in green, while all other grids are marked in red, which are defined as "risk areas" that may be invaded by illegal nodes such as spoofing attack nodes or Sybil nodes.

Because the wireless channel has the characteristics of time-variation, space-variability, uniqueness, and reciprocity, the communication channels between the transmitter and the receiver in different locations are totally different. The physical layer authentication based on wireless channel fingerprint uses the channel diversity caused by spatial variability to achieve authentication. In coherence time, the physical layer attributes are used to verify the uniqueness and reciprocity of wireless channels on both sides of the communication. In essence, it is a comparison of the channel feature consistency of adjacent packets.

Using CSI as wireless channel fingerprints to identify nodes is an important means of physical layer authentication. The uniqueness in time and space of the wireless channel allows us to map different locations with different Spatio-temporal environment characteristics. In the process of communication, the instantaneous CSI extracted during the access to AP is distinct for legal nodes and illegal nodes located in different locations. Based on the instantaneous CSI, the deep features can be further extracted to reflect the characteristics of the Spatio-temporal environment, so as to uniquely identify the sending nodes located in different positions.

We use $\vec{X}$ and $\vec{Y}$ to symbolize the signal vectors of the transmitter and receiver. MIMO channels can be modeled as:

$$\vec{Y} = \mathrm{H}\vec{X} + \vec{N}, \tag{1}$$

where $\vec{N}$ is additive white Gaussian noise and CSI is the channel frequency response H, which can be estimated by $\vec{X}$ and $\vec{Y}$:

$$\mathrm{CSI} = \hat{\mathrm{H}} = \frac{\vec{Y}}{\vec{X}}. \tag{2}$$

Wireless signals from the transmitter may suffer from multipath effects, fading, shadows, and delay distortions. CSI records the channel changes experienced during signal propagation and has the following characteristics:

(i) Due to multipath effects and channel fading in the environment, the instantaneous CSI data obtained at different locations are different, and authentication based on CSI deep characteristics can achieve resolution at a precision of centimeter-level [11].

(ii) Compared with RSS and other physical layer attributes that reflect large-scale fading in the channel, CSI contains more detailed location information and mirrors deeper channel differences.

In order to prevent spoofing attacks and the Sybil attack, and then ensure the secure access of legal nodes in the IoT area, this paper proposes a physical layer authentication mechanism based on CNN, which utilizes the strong coupling relationship between instantaneous CSI and node position, thus extracting CSI deep features as node identity to realize the identification of foreign unknown nodes within the pre-set area.

III. PHYSICAL LAYER AUTHENTICATION MECHANISM BASED ON CNN

3.1 Establishment of Authentication Mechanism

The proposed authentication mechanism is shown in Figure 2. The establishment of the mechanism is composed of three stages: instantaneous CSI data acquisition; CSI deep feature extraction and authentication model establishment; and access authentication of unknown nodes.

(i) **Instantaneous CSI acquisition phase.** We divided the IoT area into N grids and sampled several times in different grids in the pre-set IoT area to obtain plenty of instantaneous CSI data of legal or illegal nodes, so as to provide sufficient training samples for

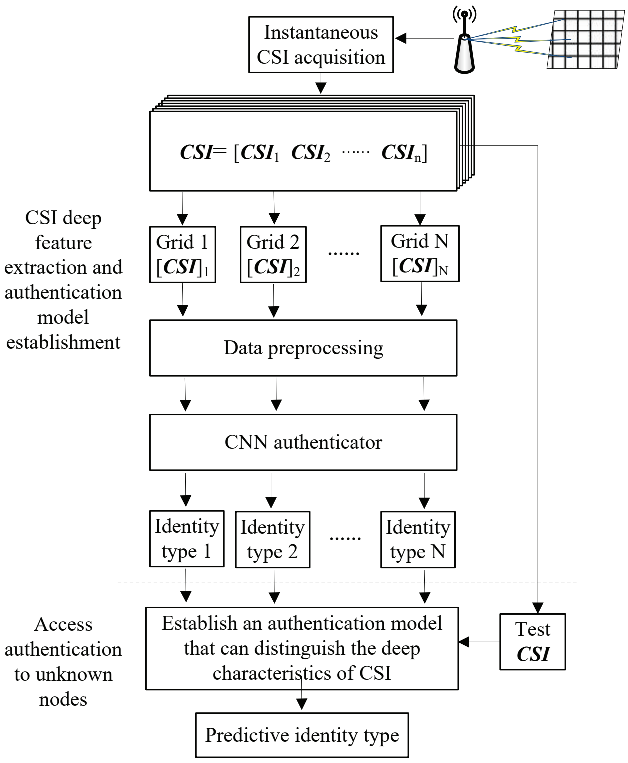


Figure 2. *Physical layer authentication mechanism based on CNN.*

model establishment. N instantaneous CSI matrices are obtained by sampling n times in N grids.

(ii) **CSI deep feature extraction and authentication model establishment phase.** The specific practices are as follows: Since the CNN authenticator requires certain data formats to work, it is necessary to preprocess the input data, including the normalization of instantaneous CSI data and the tagging of the CSI matrix. Normalization means that the input values are mapped to the range of [0, 1] to improve the learning efficiency of CNN authenticators, while tagging means that N instantaneous CSI matrices are preprocessed and different identity tags are added to distinguish the instantaneous CSI information of nodes located in different grids. In addition, before using the CNN authenticator to identity IoT nodes, it is indispensable to train the CNN authenticator. Specially, we fed the preprocessed data set into the CNN authenticator for training to extract CSI deep features, so as to obtain an authentication model that can represent and distinguish Spatio-temporal environment features in different locations.

(iii) **Unknown nodes access authentication phase.**

Algorithm 1. *CNN-based physical layer authentication algorithm.*

```
Input: CSI data set Ω
  Normalize the data set to get Ω′
  Add label i to the data set, i ∈ N ∩ i ∈ [0, 99]
  Input Ω′ and labels into the CNN network for training
  for new CSI do
    Extract CSI deep features
    Get the predictive value Ĉt with trained authentication model, Ĉt ∈ N ∩ Ĉt ∈ [0, 99]
    if Ĉt ∈ [0, 9] then
      Judging that the identity of the unknown node is legal
    else
      Judging that the identity of the unknown node is illegal
    end if
  end for
Output: Identity type of the unknown node (Alice1-10 or Eve1-90)
```

When the unknown node breaks into the area and sends the access request to AP, instantaneous CSI data are collected and fed into the fully trained authentication model to extract the CSI deep features, so that the legitimacy of the unknown node's identity can be detected efficiently. The mathematical expression at this stage is shown in formula 3:

$$\hat{C}_{\mathrm{t}} = f(H(t)), \tag{3}$$

where $\hat{C}_{\mathrm{t}}$ represents the authentication identity type of the output. We defined 100 identity types totally, including 10 legal identity categories (Alice1-Alice10) and 90 illegal identity categories (Eve1-Eve90). $f(\cdot)$ is the physical layer authentication mechanism proposed in this paper, which can effectively distinguish the user identity type, $H(t)$ represents the channel observation at the moment.

The establishment process of the proposed authentication mechanism is shown in Algorithm 1.

3.2 Design of CNN Authenticator

CNN authenticator is the core of the proposed mechanism, and its design has a significant impact on the

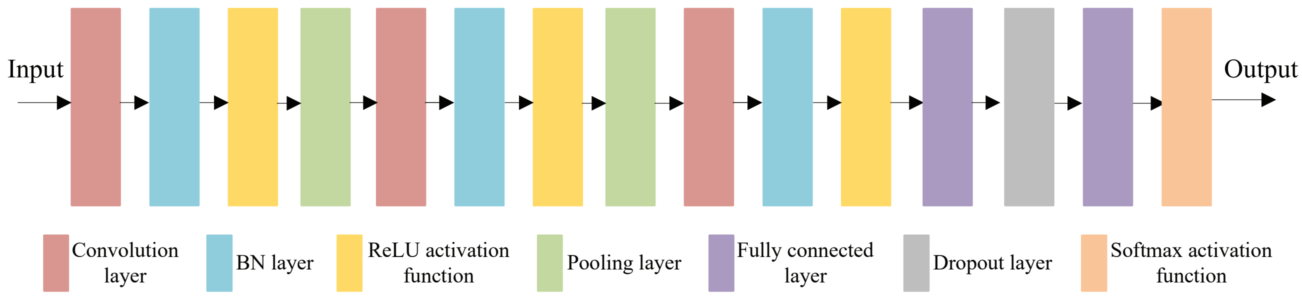


Figure 3. *The structure of CNN authenticator.*

authentication performance of the overall facility. The structure is shown in Figure 3.

The proposed CNN authenticator consists of three convolution layers, two pooling layers, and two full connection layers. Each convolution layer is followed by a batch normalization (BN) layer and a ReLU activation function. In the process of network training, the BN layer can normalize the same dimensional characteristics of a batch of samples, thus accelerating the model training and improving the training accuracy. The activation function provides the nonlinear modeling ability of the network, which makes it can nonlinearly transform the separable CSI data so that the given authenticator is able to capture and model potential core deep features.

However, sampling errors or insufficient training data may lead to over-fitting, which means the prediction accuracy may be unsound in the test stage even though the model performs well in the training stage. In order to prevent over-fitting, a dropout layer is introduced between the two fully connected layers. After the last fully connected layer, an output layer activated by the softmax function is connected. The softmax function normalizes the output results and maps all the output values to the range of [0, 1], thus judging the identity types corresponding to the input node data.

As can be seen from Figure 3, the convolution layer, pooling layer, and fully connected layer are indispensable components of CNN authenticator, which are described in detail below.

(i) **Convolution layer.** The convolution layer is the core of the CNN authenticator. Through multiple convolution, the input can be divided into various "small areas" as the input of the fully connected layer, so as to find a reasonable indirect connection mode and reduce the input parameters of the fully connected layer. In the convolution layer, the convolution calculation of data is realized by the convolution kernel. The convolution calculation of each neuron in the convolution layer is given by formula 4.

$$\begin{aligned} & a^{(l+1)}(i,j) \\ & = f_l([a^{(l)} \otimes w^l(i,j) + \Xi]) \\ & = f_l(\sum_{k=1}^{K_l}\sum_{x=1}^{g_l}\sum_{y=1}^{g_l}[a_k^{(l)}(s_l \cdot i + x, s_l \cdot j + y)w_k^l(x,y)] + \Xi), \end{aligned} \tag{4}$$

where $a^{(l)}$ and $f_l(\cdot)$ represent the output and the activation function of the neurons in layer l respectively; $n_l \times n_l$ represents the output dimension of the neurons in layer l, while $m_l \times m_l$ represents the size of the convolution kernel in layer l; K_l and g_l severally represent the number of convolution kernels and neurons in layer l; w_k^l means the weight parameter of k convolution kernels in layer l; s_l and p_l individually represent the convolution step and filling size in the convolution layer l; Ξ represents the threshold matrix. Filling means that the convolution kernel fills 0 element, 1 element, or the same element (i.e, 0 filling, 1 filling, and same filling) around the input matrix during the convolution operation, so that the dimension of the convoluted output matrix is consistent with that of the input matrix, and the step size specifies the fixed distance for each movement of the convolution kernel.

(ii) **Pooling layer.** The pooling layer uses the overall characteristics of the area to replace the network output. The pooling layer can realize down-sampling, which can reduce the matrix dimension, the amount of computation, and the network complexity by partitioning the data. The common pooling methods are maximum pooling and average pooling, where the maximum or average value in the area is calculated as the pooled value. The calculation methods of maximum pooling and average pooling can be expressed by formulas 5 and formulas 6, respectively:

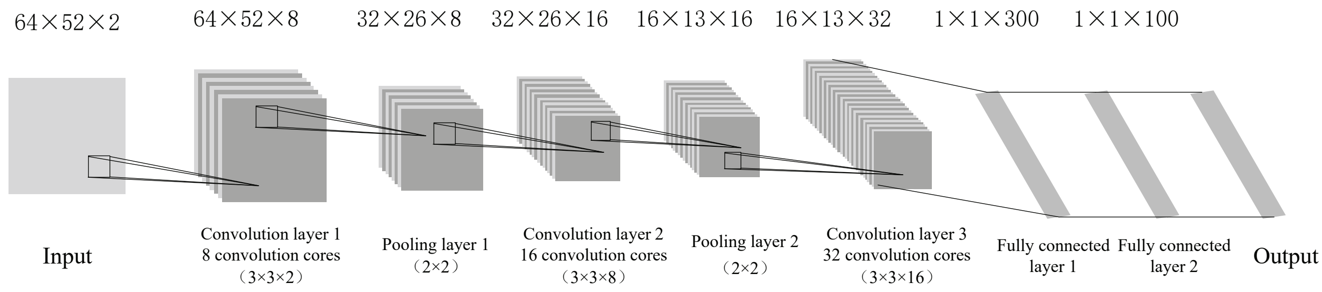


Figure 4. *CNN parameter setting.*

$$a^{(l+1)}(i,j) = \max_{k,x,y}(a_k^l(s_l \cdot i + x, s_l \cdot j + y)), \quad (5)$$

$$a^{(l+1)}(i,j) = \frac{1}{K_l f_l f_l} \sum_{k=1}^{K_l} \sum_{x=1}^{g_l} \sum_{y=1}^{g_l} [a_k^{(l)}(s_l \cdot i + x, s_l \cdot j + y)]. \quad (6)$$

(iii) **Fully connected layer.** The fully connected layer connects all the elements, merges the local features into global features, and sends the output values to the classifier to classify the input.

IV. EXPERIMENTAL DESIGN AND RESULT ANALYSIS

4.1 Acquisition of Instantaneous CSI

In this study, Quasi-Deterministic Radio Channel Generator (QuaDRiGa) [12] simulation platform, which is developed from the WINNER channel model, is used to simulate the channel environment and obtain CSI data. Compared with the WINNER II [13] channel model, QuaDRiGa has more detailed Input parameters for geometry and topology of the simulation scene and characteristics of continuous-time evolution and 3D MIMO. In addition, the Drifting model is introduced into the QuaDRiGa simulation platform, which can realize the smooth evolution of small-scale parameters such as power, delay, emission angle, and arrival angle of multipath in a short time interval when the mobile terminal moves within a segment. Li et al. [14] verified the accuracy of large-scale parameter statistics and time evolution characteristics of QuaDRiGa simulation platform based on measured data.

In this experiment, we considered an area of 20 m × 20 m, in which the position of the legal node is fixed and the AP is located 5 meters above the center of the area. We divided this area into 100 grids with the size of 2m × 2 m and sampled 500 times in each grid. The CSI data in each grid is obtained by AP and incorporated into the data set. The specific simulation parameters are set as shown in Table 1 to control sequence.

Table 1. *Simulation parameter settings.*

Options	Parameter Settings
Channel Model	3GPP 38.901 UMa NLOS
Center Frequency	2.4 GHz
Bandwidth	3 MHz
Number of Subcarriers	52
Polarization Direction	Vertical Polarization
Antenna Configuration	AP: 64 Antennas Nodes: Single Antenna

Through the QuaDRiGa channel simulation platform, a 64×52×50000 CSI complex matrix can be obtained. The real part and imaginary part of the complex matrix are split and merged into a 64×52×2×50000 four-dimensional matrix, which can be easily used as the input of CNN.

4.2 Data Preprocessing

Before feeding the data into CNN, it is necessary to normalize the data. We carried out the linear transformation of the original data with the maximum-minimum normalization method, limiting the data to the range of [0,1], which can improve the learning efficiency and accelerate the convergence speed. The maximum-minimum normalized mathematical expression is shown in formula 7.

$$y = \frac{(x - \max)}{(\max - \min)}. \quad (7)$$

After normalization, the data needs to be tagged. Since one category is supposed to be mapped with the CSI data in the one grid, there are 100 categories that correspond to data in 100 grids, including 10 legal identity categories (Alice1-Alice10) and 90 illegal identity categories (Eve1-Eve90). In this way, after identifying the identity of the unknown node, the grid location of the unknown node can be determined directly according to its identity type.

4.3 CNN Network and Set of Training Parameters

CNN uses back propagation algorithm [15] in training. Firstly, the activation output of each layer is obtained by forward propagation, and when updating the parameters, the gradient descent algorithm (such as adam algorithm, sgdm algorithm, rmsprop algorithm [15], etc.) is used to update the parameters from the last layer forward in turn, thus realizing back propagation. The CNN network parameters are set as shown in Figure 4.

Same filling is used in each convolution layer and the step size is set as 1, while the maximum pooling method is utilized in the pooling layer and the step size is set as 2. Figure 4 shows the matrix dimension of the output after the data enters each layer. The input dimension is 64×52×2. In convolution layer 1, 8 convolution kernels with the size of 3×3×2 perform convolution operations on the input data, and then the output with a dimension size of 64×52×8 is obtained through normalization and nonlinear activation. After reaching the pooling layer 1, the pooling layer reduces the dimension of the data and extracts the main features, and the output dimension is 32×26×8. After that, the data reaches the full connection layer through convolution layer 2, pooling layer 2, and convolution layer 3, finally the output with a dimension of 1×1×100 is obtained.

The four-dimensional matrix data set of 62×52×2×50000 is divided into training set, verification set, and test set at the proportion of 8:1:1, and the adam algorithm is used for training. Several important training parameters are set as shown in Table 2, where Batch refers to the size of samples selected in the training set to update weights; iteration refers to the process of updating weights with a batch; an epoch refers to complete training of all data in the training set. The initial learning rate is an important parameter in training, and the learning rate afterward determines the efficiency of the training. The training will take a long time if the learning rate is too low, but if the learning rate is too high, the training may fall into sub-optimal results. The learning rate can be adjusted adaptively in the training process. With the increase of epoch in the training process, the training result is approximating to the optimal value, and then the learning rate can be appropriately reduced to continue the training process. Learn rate drop factor is a multiplier factor used to adjust the learning rate while learn rate drop period decided to adjust the learning rate after several epoch. In this experiment, it is set to reduce the learning rate to half after every 5 epoch. Validation frequency is used to determine how many times to use the validation set to validate the model after iteration.

Table 2. *Training parameter setting.*

Parameters	Numerical Value
Mini Batch Size	200
Iteration	200
Epoch	10
Initial Learn Rate	0.001
Learn Rate Drop Factor	0.5
Learn Rate Drop Period	5
Validation Frequency	200

4.4 Analysis of Simulation Results

4.4.1 Performance analysis

The authentication performance of the physical layer authentication mechanism based on CNN proposed in this paper is shown in Figure 5. The verification accuracy and training accuracy increase with the expansion of epoch and the number of iterations, and finally tend to be stable. At the end of the training process, the test set is used to test the training model, and the test accuracy is calculated by formula 8.

$$accuracy_{test} = \frac{sum(Y_{pred} = Y)}{num_{test}}, \tag{8}$$

where Y_{pred} and Y are the prediction result and the real result, respectively. num_{test} is the data quantity of the test set. The test accuracy is the proportion of

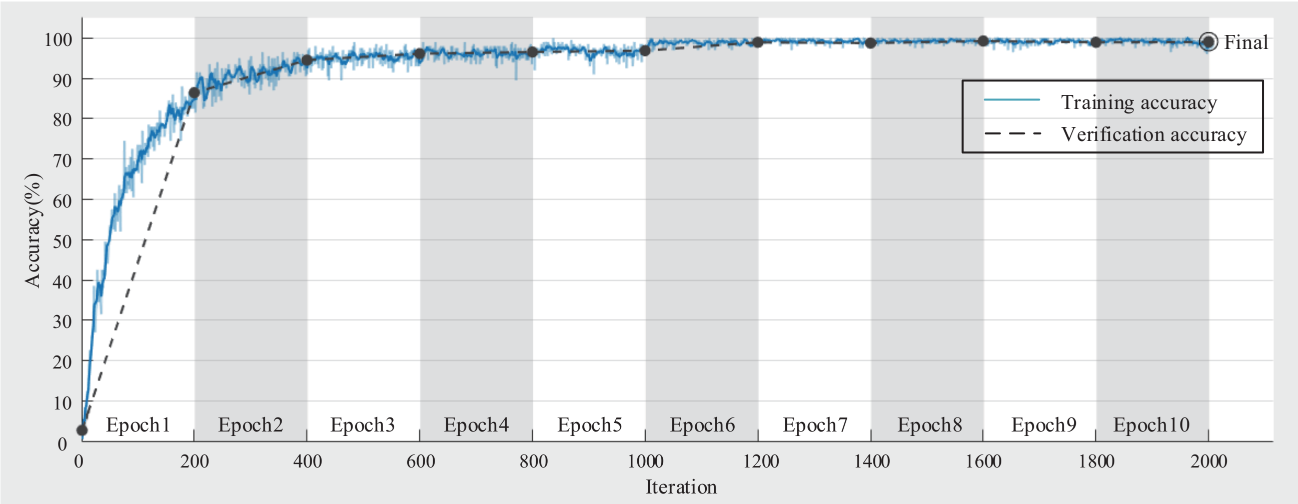


Figure 5. *Authentication accuracy.*

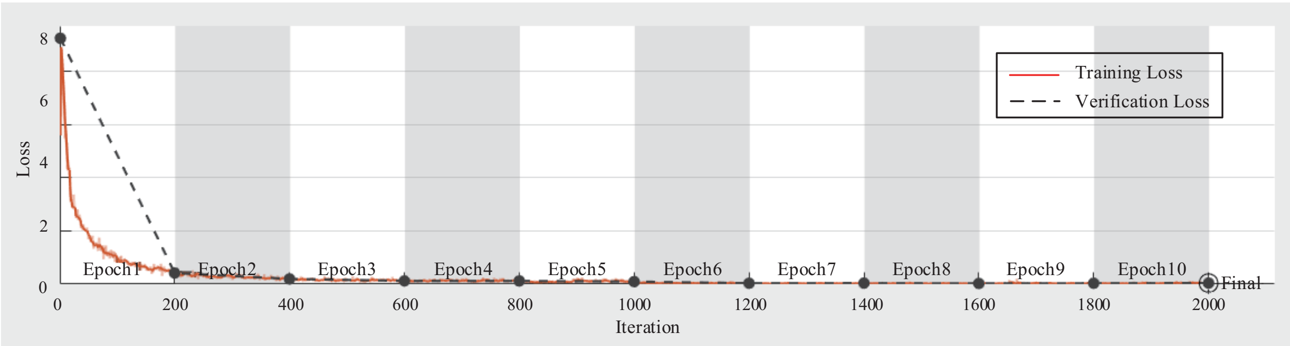


Figure 6. *Change of CNN loss function.*

successful prediction result to the num_{test}, and the final test accuracy is 99.52%.

The loss function reflects the fitting degree of the training model to the data, which means the gap between the predicted value and the real value of the model. When the deep learning network is used to deal with the classification problem, the training and verification loss are usually measured by the cross-entropy loss function. The smaller the loss value is, the more accurate the model is. The cross-entropy loss function can be expressed by formula 9:

$$C = \frac{1}{n}\sum_{i=1}^{n}[Y\ln(Y_{pred}) + (1-Y)\ln(1-Y_{pred})]. \tag{9}$$

Figure 6. shows the change of the CNN loss function. It can be seen that the training loss and verification loss gradually converge to zero with the increase of epoch and the number of iterations, and the training model fits the data well.

The above analysis shows that the proposed mechanism in this paper can achieve high authentication accuracy. Furthermore, after the model training, it can quickly identify the unknown node and locate its grid region according to the classification prediction results, so as to realize the real-time detection of illegal intrusion nodes.

In order to demonstrate the robustness of the proposed authentication mechanism, we mix Gaussian white noise into the extracted CSI data to simulate the channel estimation error generated during channel estimation, the test accuracy obtained under different

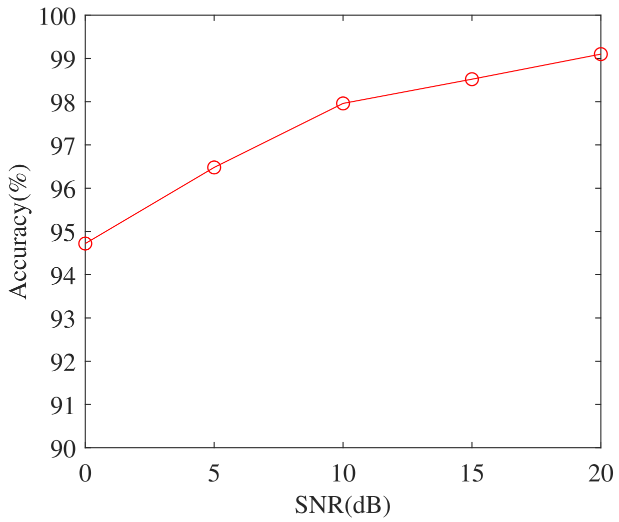
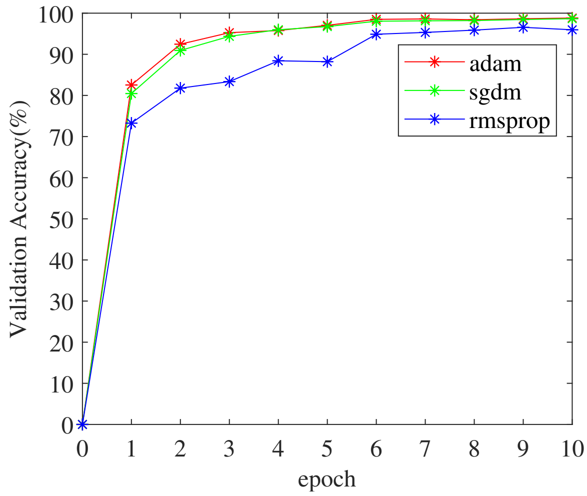
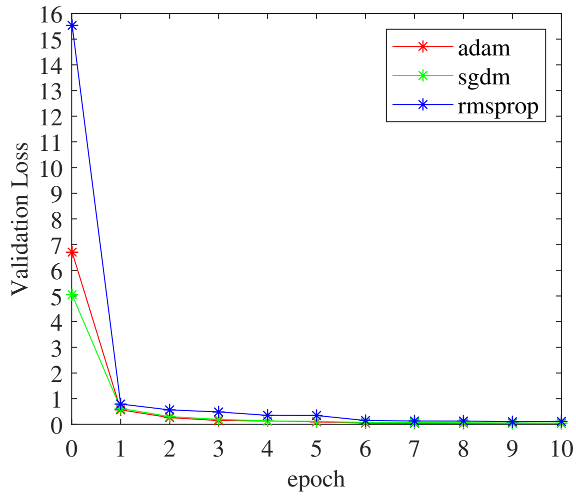


Figure 7. *CNN training accuracy under different SNR.*



(a) *verification accuracy.*



(b) *verification loss.*

Figure 8. *CNN verification accuracy and verification loss using different gradient descent algorithms.*

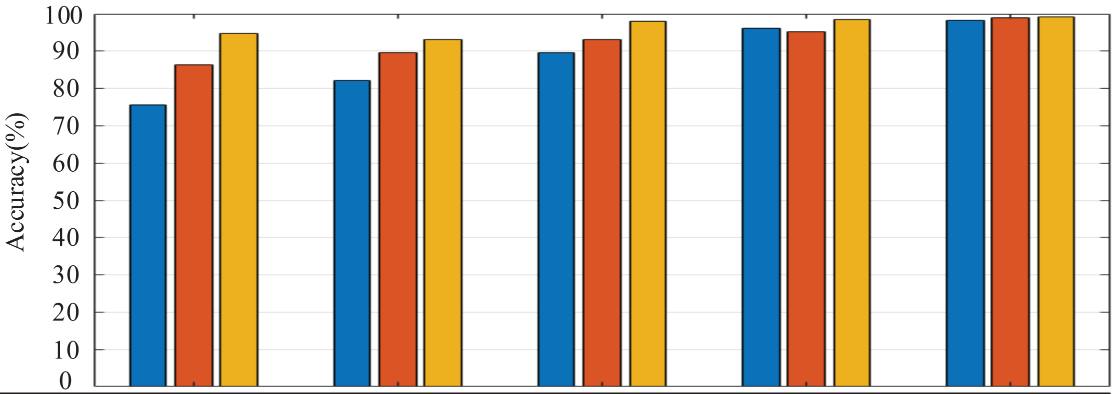
SNR (e.g, 0 dB, 5 dB, 10 dB, 15 dB, 20 dB) is shown in Figure 7. With the increase of SNR, the signal features are easier to extract, and the test accuracy keeps elevating.

Different gradient descent algorithms will produce different training results. Figure 8. compares the verification accuracy and verification loss when using adam algorithm, sgdm algorithm, and rmsprop algorithm respectively. The rmsprop algorithm adaptively retains the learning rate for each parameter based on the average value of the nearest magnitude of the weight gradient. The sgdm algorithm adds inertia in the process of gradient descent, which makes the effects in two aspects. First, the updating speed becomes faster in the dimension where the gradient direction is unchanged, while the updating speed changed slower in the dimension with the varying gradient direction so that it can accelerate the convergence and reduce the concussion.

Adam is a first-order optimization algorithm that can replace the traditional random gradient descent process. It combines the advantages of adaptive gradient algorithm and rmsprop algorithm, which makes it can iteratively update the weights of neural networks based on training data. The experiment shows that the performance of adam algorithm and sgdm algorithm is slightly better than that of rmsprop algorithm.

4.4.2 Comparison with traditional schemes

In this part, the performance of the proposed authentication mechanism is compared with the traditional authentication scheme. Xiao L et al [16] proposed a physical layer authentication method based on the hypothesis test for spoofing attacks in wireless networks. Specifically, they compared the test statistics to the authentication threshold to determine whether the user identity is legal or illegal based on LS channel estimation. Zhang et al. [17] proposed a method based on basis extension model (BEM) to improve the performance of physical layer authentication. The method approximates the state of the separable path by using



SNR / Scheme	0dB	5dB	10dB	15dB	20dB
LS	75.6%	82.1%	89.5%	96.2%	98.2%
BEM	86.3%	89.6%	93.1%	95.2%	99.0%
CNN	94.7%	96.5%	97.9%	98.5%	99.1%

Figure 9. *Comparison of authentication accuracy between physical layer authentication mechanism based on CNN and traditional authentication scheme.*

the mutual orthogonal basis function and the constant base coefficient, so as to transform the correlation between the sampling values of the separable path into a resource within the block transmission time and enhance the security authentication mechanism.

Figure 9. shows the authentication accuracy of the proposed mechanism and the above two traditional schemes under different SNR. It can be seen that under the condition of high SNR (e.g, 15 and 20 dB), the three authentication schemes can achieve more than 95% authentication accuracy. However, in the case of low SNR (e.g, 0 and 5 dB), the authentication accuracy of the authentication scheme based on LS channel estimation and BEM is less than 90%, while the physical layer authentication mechanism based on CNN proposed in this paper maintains the authentication accuracy of up to 94%, which is obviously better than the traditional authentication scheme based on hypothesis test and BEM.

4.4.3 Comparison with the deep learning scheme

Furthermore, we compare our mechanism with the other one physical layer authentication method based on deep learning proposed by R. Liao, et al [18].

R. Liao, et al. exploited CSI to enhance MIMO-OFDM system security by detecting spoofing attacks in wireless networks. Otherwise, they considered multiple legitimate transmitters and one spoofing attacker. We have compared our scheme with it in both the network model and the authentication accuracy, and the results are shown in Table 3. Compared with their scheme, we have considered the network model and

Table 3. *Comparison in network model and authentication accuracy.*

	Network Model	Accuracy (under 4dB)
Our Method	Conv1 3×3×2-8 Maxpooling Conv2 3×3×8-16 Maxpooling Conv2 3×3×16-32 FC1 300×1 FC2 100×1	96.1%
Method Proposed by R Liao	Conv1 4×4×1-8 Maxpooling Conv2 2×2×8-16 Maxpooling FC 256×1	90%

parameter settings more carefully, it can be seen from the results that our proposed scheme is more stable and has a certain degree of anti-noise ability.

V. CONCLUSION

In order to ensure the secure access of pre-set IoT nodes within the area and further improve the authentication accuracy, we propose a physical layer authentication mechanism based on CNN. Specifically, we extract CSI deep features as wireless channel fingerprints to distinguish sending nodes in different locations. The establishment of the authentication mechanism is divided into three stages: instantaneous CSI acquisition, CSI deep feature extraction and authentication model establishment, and unknown node access authentication. First, the instantaneous CSI information collected in different grids is fed into the CNN authenticator to extract CSI deep features, then a robust authentication model is established to identify unknown nodes and further determine their location, thus realizing the IoT regional intrusion detection finally. We analyze the performance of the proposed authentication mechanism and compare it to the previous schemes. The experimental results show that the trained CNN-based physical layer authentication mechanism can achieve the test accuracy of 94.7% even in the case of low SNR of 0 dB, which means it obviously performs better than the traditional authentication scheme and is more stable.

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